



Degree, Semester & Branch: I Semester B.E. CSE – B – Section

Course Code & Title: GE3151 Problem solving and python programming

Name of the Faculty member (s): Dr.R.Venkatesh

Innovative Practice Description

Unit / Topic: Unit I / Illustrative problem: Towers of Hanoi

Course Outcome: CO 1

Topic Learning Outcome: TLO 2,3

Activity Chosen: Role Play

Objective:

The objective of the role-play activity was to engage students in solving the "Recursion and Towers of Hanoi" problem through interactive learning, which would reinforce their understanding of recursion and problem-solving strategies in Python programming.

Time Allotted for the Activity: 15 minutes

Roles Assigned:

- **Student 1:** Role of the "Solver" (Karthick S), who moves the disks following the Towers of Hanoi rules.
- **Student 2:** Role of the "Move Counter" (Mukesh Kalyan K), responsible for counting and verifying the number of moves made by the Solver.

Details of the Implementation:

1. Introduction to Problem:

The instructor explained the "Towers of Hanoi" problem, its recursive solution, and its relevance to problem-solving and algorithms in Python. The rules of the game were presented to the class, which involves moving disks from one rod to another following specific constraints.

2. Role Assignment:

- **Student 1** was given the task of physically solving the problem by moving disks from the first rod to the third rod.
- **Student 2** kept track of the number of moves and interacted with Student 1 to ensure the correct number of moves were followed.

3. Activity Execution:

- **Student 1** started moving the disks while abiding by the rules:
 - Only one disk can be moved at a time.
 - A disk can only be placed on top of a larger disk or an empty rod.
- Student 2 kept count and announced when the solution approached the optimal number of moves ($2^N - 1$).
- The rest of the class observed the activity and provided feedback, offering suggestions for optimizing the number of moves.

4. Completion:

The task was completed after several attempts. Student 1 managed to solve the Towers of Hanoi problem in 7 moves for a 3-disk problem, adhering to the optimal solution.

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PSO1
CO 1	2	1	1	1

(1 – Low 2 – Moderate 3 – High)

PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PSO1
	2	1	1	1
Justification for correlation	Apply basics mathematics knowledge to solve the problem	Solutions can be identified, formulated to solve the problem	Solution for the problem can be drawn using flow chart	Helped the students in solving various problems in real world

• Images / Screenshot of the practice:



Figure 1: Role Play Activity

Learning Outcome:

• Understanding of Recursion:

The role-play helped students visualize the recursive nature of the Towers of Hanoi problem. They could see how each smaller sub problem relates to the overall solution.

• Problem-Solving Skills:

The activity allowed students to practically apply the recursive strategy learned in theory, enhancing their problem-solving capabilities.

• Engagement and Interaction:

The hands-on experience encouraged even passive learners to engage in the lesson. Active participation promoted a better understanding of the topic.

Reflective Critique:

- **Student Feedback:**

Students reported that the activity made the Towers of Hanoi problem much easier to understand. The role-play format allowed them to visualize and internalize the recursive steps needed to solve the problem. It also made writing the algorithm and drawing flowcharts more straightforward.

- **Classroom Dynamics:**

The activity promoted collaboration among students. Watching their peers actively solve the problem inspired other students to participate and ask questions.

- **Benefits of the Practice:**

- ✓ Enhanced retention of the topic through practical experience.
- ✓ Improved understanding of recursion and its application in programming.
- ✓ Created a dynamic and interactive classroom environment.

Challenges Faced:**Limited Participation:**

The activity involved only two students, which left the rest of the class as observers. It was challenging to ensure that all students could engage in hands-on learning.

Time Constraint:

Completing the activity within the allocated time was difficult, especially with students taking multiple attempts to solve the problem optimally.

Conclusion:

The role-play activity on the Towers of Hanoi problem was a success in terms of improving student engagement, enhancing their understanding of recursion, and making learning more interactive. While the activity involved only two students directly, the entire class benefited from observing and discussing the steps involved. Future sessions can include more students in similar activities to maximize involvement.

References:

- <https://serc.carleton.edu/introgeo/roleplaying/whatis.html>
- https://www.ritrjpm.ac.in/images/computer-science/10_CS8591_Rolepaly.pdf
- <https://www.niu.edu/citl/resources/guides/instructional-guide/role-playing.shtml>

Signature of Faculty Member

HOD-CSE